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Appendix A. Proof of Theorem 3.9.

In this section, we prove Theorem 3.9.

Define S as

$$(A.1) \quad S := \left\{ \{b_i\}_{i=1}^{\infty} \mid 0 \leq b_i \leq a_i, \sum_{i=1}^{\infty} b_i = 1 \right\},$$

where $\{a_i\}_{i=1}^{\infty}$ is a sequence defined in Assumption 3.6. Then, the following lemma holds.

LEMMA A.1. S is a compact subset of $\ell^1(\mathbb{N})$.

Proof. Let $\{\beta^{(n)}\}_{n=1}^{\infty}$ be a sequence in S , where $\beta^{(n)} := \{b_i^{(n)}\}_{i=1}^{\infty} \in S$. Since $0 \leq b_1^{(n)} \leq a_1$ holds for any $n \in \mathbb{N}$, the sequence $\{b_1^{(n)}\}_{n=1}^{\infty}$ has a convergent subsequence $\{b_1^{(1_n)}\}_{n=1}^{\infty}$. Similarly, since $0 \leq b_2^{(1_n)} \leq a_2$, $\{b_2^{(1_n)}\}_{n=1}^{\infty}$ also has a convergent subsequence $\{b_2^{(2_n)}\}_{n=1}^{\infty}$. Repeating this procedure for each coordinate and applying the diagonal argument, we obtain a subsequence $\{\beta^{(n_n)}\}_{n=1}^{\infty}$. For all $i \in \mathbb{N}$, the sequence $\{b_i^{(n_n)}\}_{n=1}^{\infty}$ converges, and $0 \leq b_i^{(n_n)} \leq a_i$ holds. Define $c_i := \lim_{n \rightarrow \infty} b_i^{(n_n)}$. Since $0 \leq b_i^{(n_n)} \leq a_i$, it follows that $|b_i^{(n_n)} - c_i| \leq a_i$. Moreover, since $\sum_{i=1}^{\infty} a_i < \infty$, we can apply Lebesgue convergence theorem. This yields $\lim_{n \rightarrow \infty} \sum_{i=1}^{\infty} |b_i^{(n_n)} - c_i| = 0$, which implies that the subsequence $\{\beta^{(n_n)}\}_{n=1}^{\infty}$ converges to c with respect to ℓ^1 norm. Furthermore, it is straightforward to verify that $\{c_i\}_{i=1}^{\infty} \in S$. Thus, $\{\beta^{(n)}\}_{n=1}^{\infty}$ has a subsequence that converges in S . Therefore, S is compact. $\square$

LEMMA A.2. Under Assumption 3.6, the feasible region $\mathcal{F}$ is defined as

$$(A.2) \quad \mathcal{F} := S \cap X,$$

where

$$(A.3) \quad X = \{\{p_i\}_{i=1}^{\infty} \mid \mu_n(W(p)) > 0\}.$$

Then $\mathcal{F}$ is a convex set.

Proof. Since the convexity of S is straightforward, we show the convexity of X . Then, it suffices to show that $\mu_n(W((1-\theta)p + \theta q)) > 0$ for any $\theta \in (0, 1)$ and $p, q \in X$.